

# Point Cloud Semantic Segmentation with Sparse Convolutions

MinkUNet on SemanticPOSS, Lab 4 Scientific Report

3D Data Processing 2025/2026, University of Padova (DEI)

Giuseppe D'Auria, Student ID 2163508

## 1. Introduction and Experimental Setup

I implement MinkUNet, a sparse 3D U-Net built on the spconv library, for point-level semantic segmentation on a reduced SemanticPOSS dataset with 13 valid classes and ignore index 255. The pipeline voxelizes each LiDAR scan, averages point features inside every active voxel, processes the sparse grid with submanifold and strided sparse convolutions, restores resolution through inverse sparse convolutions with skip connections, and projects voxel predictions back to the original points through an inverse coordinate map before computing mIoU.

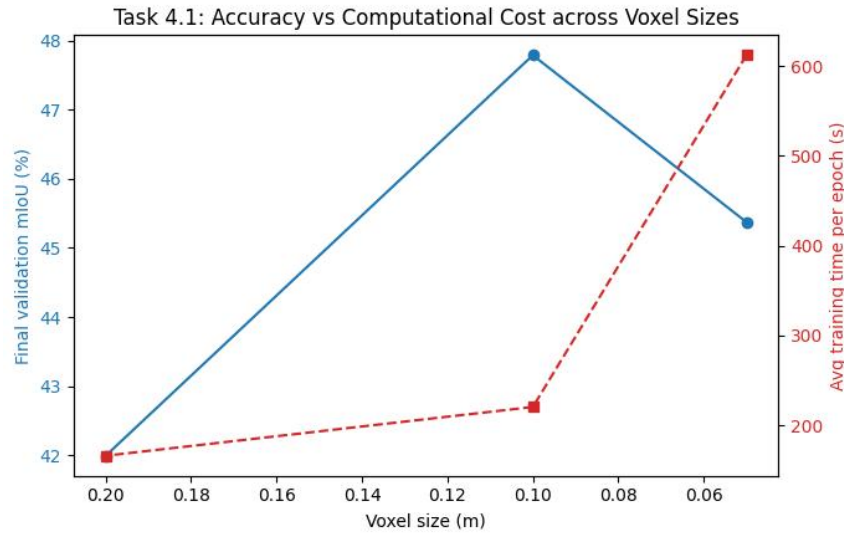
**Common configuration.** AdamW with learning rate  $1e-3$  and weight decay  $1e-4$ , cross-entropy loss with ignore\_index 255, training on sequence 0, validation on sequence 1, test on sequence 2, 5 epochs per run. mIoU is computed at the point level after the inverse mapping, with absent classes excluded from the mean.

**Note on runtime measurement.** The three voxel-size runs in Task 4.1 used per configuration batch sizes of 2, 4 and 8 for voxel sizes 0.05, 0.10 and 0.20 m respectively, because the finest grid does not fit the available 16 GB GPU at a large batch. The reported per-epoch times therefore reflect both the voxel resolution and the batch size, and the runtime column should be read as an operating-point comparison rather than a single-variable isolation of voxel size. The accuracy column is also influenced by batch size, not only by voxel size: with a fixed training set the number of gradient updates per epoch equals the dataset size divided by the batch size, so over the same 5 epochs the batch 8 run performs roughly four times fewer updates than the batch 2 run, and batch normalization statistics and the learning rate to batch size interaction differ as well. The accuracy figures should therefore also be read as a joint voxel size and batch size operating point rather than a single variable isolation of voxel size. The two Task 4.2 runs share an identical batch size of 4, so that comparison is fully controlled.

## 2. Task 4.1, Voxel Size Sensitivity Study

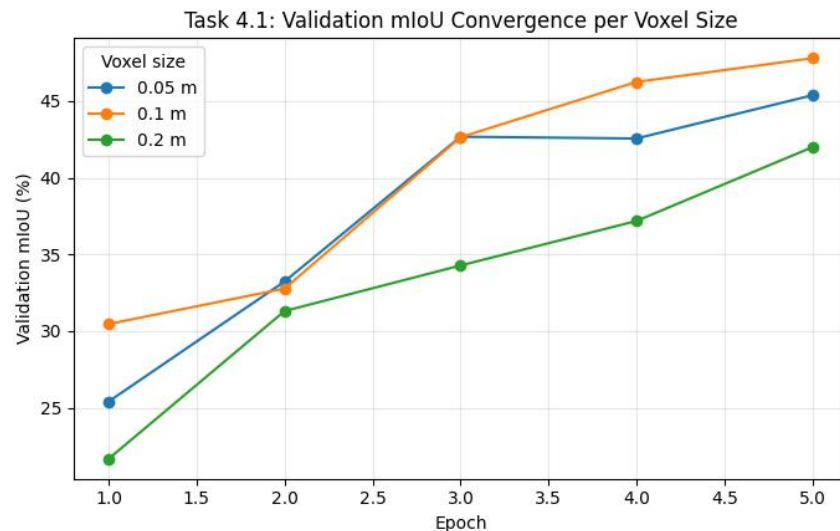
### 2.1 Results

| Voxel size (m) | Batch | Avg epoch time (s) | Best val mIoU over 5 epochs (%) |
|----------------|-------|--------------------|---------------------------------|
| 0.05           | 2     | 612.04             | 45.37                           |
| <b>0.10</b>    | 4     | 220.56             | <b>47.79</b>                    |
| 0.20           | 8     | 166.27             | 41.99                           |



**Figure 1.** Accuracy against computational cost across voxel sizes. The x axis is inverted so resolution increases to the right. Validation mIoU (blue, left axis) peaks at 0.10 m, while average epoch time (red, right axis) rises sharply toward the finest 0.05 m grid.

The best validation accuracy is reached at 0.10 m (47.79%). The finer 0.05 m grid is both the slowest, at 612 s per epoch, and slightly less accurate (45.37%), while the coarse 0.20 m grid is the fastest, at 166 s per epoch, and the least accurate (41.99%). Epoch time grows steeply as the voxel shrinks. Two effects combine: the number of active voxels per scan rises sharply, scaling closer to the inverse square of the edge length for surface-dominated LiDAR rather than the inverse cube of a dense grid, and the smaller batch used at finer resolutions increases the number of gradient updates per epoch. The accuracy column reports the best validation mIoU over the 5 epochs, which Figure 2 shows coincides with the epoch 5 value for all three runs.



**Figure 2.** Validation mIoU per epoch for each voxel size. The 0.10 m run (orange) leads throughout and keeps improving to epoch 5, whereas the 0.05 m run (blue) is flat from epoch 3 to 4 and then climbs to its epoch 5 maximum, and the 0.20 m run (green) trails at every epoch.

The convergence curves add detail to the table. The 0.05 m and 0.10 m runs track each other for the first three epochs. The 0.05 m curve is briefly flat from epoch 3 to 4 and then rises to its

epoch 5 maximum of 45.37%, the steepest late climb of the three runs, while the 0.10 m curve rises more smoothly to 47.79%. Because the finer grid is still improving at the budget boundary, its deficit is most consistent with 5 epochs being insufficient for the larger and sparser active set rather than a converged limitation, and the ranking could change with more training.

## 2.2 Why too large a voxel damages accuracy

Voxelization is a lossy quantization. Every point inside a cell collapses to one averaged feature vector and one label taken from the first point. As the cell grows, two failures compound. First, geometric detail below the cell scale is erased, so thin or small structures lose the shape cues the network needs. Second, a single coarse cell increasingly straddles a class boundary and absorbs points of several classes while being forced to carry one label, which systematically penalizes small and boundary classes. This is visible in the drop from 47.79% at 0.10 m to 41.99% at 0.20 m. The effect is worst far from the sensor, where LiDAR returns are already sparse, so a large distant cell may contain only one or two points while still discarding their spatial structure.

## 2.3 Why too small a voxel is costly and does not help here

Going from 0.20 to 0.05 m the edge shrinks fourfold, so for surface-dominated data the active voxel count per scan rises by roughly sixteen times, the inverse square of that factor, while the smaller batch raises the gradient steps per epoch by about four. These factors do not multiply naively, because sparse convolution amortizes the per-voxel cost across the GPU, so the measured epoch time rises by about four times rather than by their product. Beyond cost, the 0.05 m model does not improve on 0.10 m and is in fact slightly worse. Two factors explain this. An over-fine grid fragments each surface across many one-point voxels, so the per-voxel averaging that should denoise features instead returns near raw, noisy values, and the receptive field of each layer covers less physical space. In addition, the larger and sparser active set is harder to optimize within the fixed 5-epoch budget. The points far from the sensor, where density is lowest, are where an over-fine grid produces the most isolated single-point voxels.

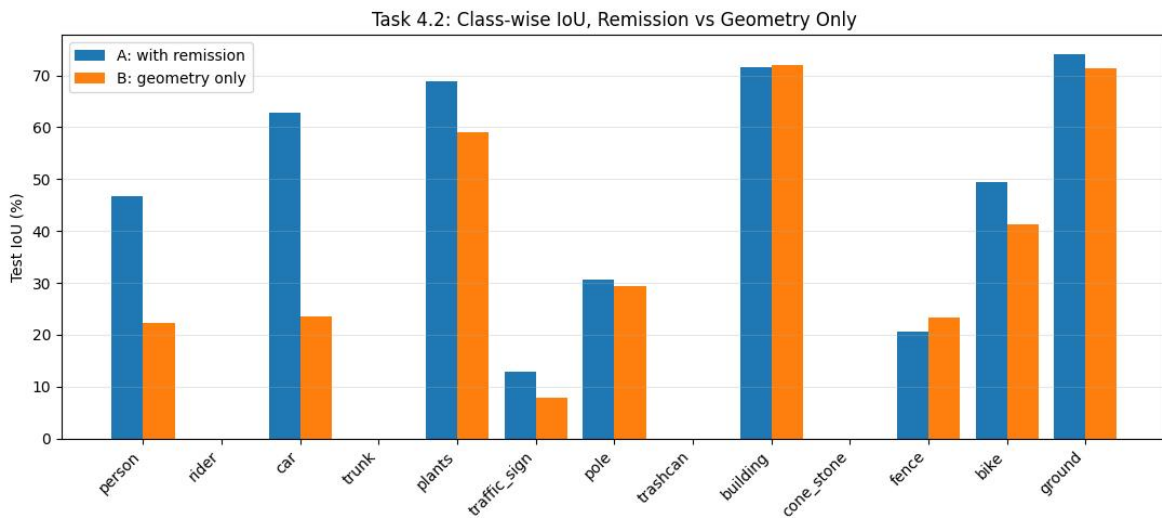
**Trade-off.** Voxel size sets the resolution versus cost operating point. The intermediate 0.10 m value gives the best accuracy at a moderate cost, which is why it is used as the fixed resolution for Task 4.2.

## 3. Task 4.2, Remission vs Geometry Only

Both models train at voxel size 0.10 m with augmentation and identical batch size 4, differing only in whether the LiDAR remission channel is kept. Model A reaches a test mIoU of 33.66%, Model B reaches 26.92%, so remission contributes 6.75 mIoU points overall. These are test figures on sequence 2, which is a different and harder split than the sequence 1 validation set used in Task 4.1, so they are lower than the 47.79% validation mIoU reported there and are not directly comparable to it. The class-wise test IoU is reported below.

### 3.1 Class-wise IoU

| Class        | A: +remission | B: geom only | delta (A-B)   |
|--------------|---------------|--------------|---------------|
| person       | 46.72         | 22.33        | <b>+24.39</b> |
| rider        | 0.00          | 0.00         | 0.00          |
| car          | 62.73         | 23.58        | <b>+39.15</b> |
| trunk        | 0.00          | 0.00         | 0.00          |
| plants       | 68.97         | 58.95        | <b>+10.02</b> |
| traffic_sign | 12.98         | 7.84         | <b>+5.14</b>  |
| pole         | 30.64         | 29.40        | +1.24         |
| trashcan     | 0.00          | 0.00         | 0.00          |
| building     | 71.52         | 71.97        | <b>-0.45</b>  |
| cone_stone   | 0.00          | 0.00         | 0.00          |
| fence        | 20.55         | 23.29        | <b>-2.74</b>  |
| bike         | 49.41         | 41.21        | <b>+8.20</b>  |
| ground       | 74.08         | 71.33        | <b>+2.75</b>  |
| <b>mIoU</b>  | <b>33.66</b>  | <b>26.92</b> | <b>+6.75</b>  |



**Figure 3.** Per-class test IoU with remission (blue) and geometry only (orange). The largest gaps appear for car, person, plants and bike. Building and pole are nearly unchanged, ground is only modestly improved, and the four zero-IoU classes are not learned within 5 epochs.

Four classes (rider, trunk, trashcan, cone\_stone) score zero IoU for both models. These are rare classes that the network does not learn to predict within 5 epochs of training on this reduced dataset, so they carry no signal for the remission comparison and are excluded from the discussion below.

### 3.2 Physical explanation of the measured results

**What remission measures.** Remission is the strength of the returned laser pulse. After the sensor range normalization it is governed largely by the target reflectivity at the LiDAR near-infrared wavelength, together with the local angle of incidence. It is a material and appearance cue that is complementary to the purely geometric coordinates.

**Vehicles and people gain the most.** The largest improvements are car at +39.15 and person at +24.39, with bike at +8.20. These classes are where geometry alone is most ambiguous after voxelization at 0.10 m, yet their materials are highly distinctive in the near infrared. Vehicles combine metal bodywork, glass and retro-reflective lights and plates, which produce a strong and characteristic intensity signature that separates them from the surrounding road and structures. People present clothing and skin reflectivity that differs from the static background and helps disambiguate a pedestrian from other thin vertical shapes such as poles. Bikes share the metallic and reflective cues of vehicles at a smaller scale.

**Vegetation.** Plants improve by +10.02. Foliage scatters and partially absorbs the near-infrared pulse, giving a diffuse low-intensity response that is distinct from the harder returns of buildings and ground, so intensity sharpens the separation of vegetation from neighbouring surfaces.

**Ground and traffic signs.** Ground improves modestly at +2.75 and traffic\_sign at +5.14. Both are materials where reflectivity is informative, asphalt versus painted markings for the ground and retro-reflective coating for signs. The gains are smaller than for vehicles because the ground is already well separated by its flat low geometry, so remission adds only a refinement, and because signs are small and rare, which keeps their absolute IoU low at 12.98% and limits the size of the measurable gain in 5 epochs.

**Geometry-dominated classes are unaffected or slightly worse.** Building changes by -0.45 and fence by -2.74, both within run-to-run noise. These structures are already well separated by shape, extent and spatial context, large planar walls and long thin barriers, so the intensity channel is largely redundant for them and can even act as a mild distractor. This contrast is the core finding: remission supplies the most value exactly where geometry is ambiguous but material is distinctive, namely vehicles, people and vegetation, and supplies little where geometry alone already determines the class.

## 4. Conclusions

- Voxel size trades resolution against cost. The intermediate 0.10 m grid gave the best validation mIoU at 47.79%, the coarse 0.20 m grid lost accuracy to quantization at 41.99%, and the fine 0.05 m grid was the most expensive at 612 s per epoch without improving accuracy, dropping slightly to 45.37%. Figure 2 shows the 0.05 m run is still rising at epoch 5, so this deficit reflects the limited 5-epoch budget for the finer, sparser grid rather than a converged result, and the ranking could change with more training.
- LiDAR remission is a complementary material cue. It raised test mIoU from 26.92% to 33.66%, with the gains concentrated in materially distinctive but geometrically ambiguous classes, car +39.15, person +24.39, plants +10.02 and bike +8.20, and negligible or slightly negative changes for geometry-defined structures such as building and fence.
- All numbers come from 5-epoch runs on a reduced dataset with a single seed, so several rare classes are not yet learned and per-class magnitudes carry run-to-run

variance. The directional findings above are nonetheless consistent and physically interpretable.